Highlights

Multi-Agent Reinforcement Learning for Surface–Subsurface Observation of Tropical Cyclones Using Submersible Unmanned Vessels

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- Dual-mode SUVs track surface winds and sample subsurface currents.
- Event-specific ERA5–HYCOM environments cover 500 historical TCs.
- Parameter-sharing IPPO is evaluated for 4- to 16-SUV fleets.
- Underwater experiments compare active, random, and passive-drift control.

Multi-Agent Reinforcement Learning for Surface–Subsurface Observation of Tropical Cyclones Using Submersible Unmanned Vessels

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ABSTRACT

Tropical cyclones remain destructive, while intensity and rapid-intensification forecasts are constrained by sparse near-core observations. Conventional platforms rarely combine rapid transit, surface-wind sampling, and subsurface-current sensing. We develop and evaluate a simulation-based multi-agent reinforcement learning framework for surface–subsurface TC observation using parameterized Blue Whale-class submersible unmanned vessels. Event-specific environments combine ERA5 10 m winds, HYCOM currents from 0–60 m, and IBTrACS tracks for 500 western North Pacific TCs. A parameter-sharing IPPO policy maps local observations to continuous commands every 10 min. Center distance is used only in training; validation tests frozen policies on unseen TCs using shared winds. Surface vehicles correct deployment errors and approach the high-wind region; underwater, they sample currents at seven depths. Ten seeds cover 4/8/12/16 vehicles at 12 and 20 m/s and validation radii of 100–800 km. Time-averaged tracking distance is lower at 20 m/s in all 20 fleet-size/radius comparisons; the fleet-averaged gap grows from 10.1 to 84.2 km. The 12 m/s scenario uses shorter paths and lower speed, revealing an efficiency–tracking trade-off. Low-pass filtering improves final tracking on average while shortening trajectories and reducing peak speed. Underwater experiments compare active IPPO, random control, and passive drift at seven depths. The surface results support active tracking; the subsurface tests quantify whether propulsion improves current sampling over drift.

1. Introduction

Tropical cyclones are among the most destructive manifestations of extreme weather. Their strong winds, extreme rainfall, storm surges, and waves threaten lives, coastal infrastructure, fisheries, shipping, and offshore operations. Recent assessments and attribution studies indicate that the global number of TCs may not increase uniformly, but the proportion of intense storms, rapid intensification events, rainfall extremes, and destructive potential has increased in many basins [1, 2, 3, 4]. Over the western North Pacific, landfalling TCs have intensified since the late 1970s, and the destructive potential of landfalling TCs over China has increased [5, 6]. Nearshore intensification and the migration of TC activity toward higher latitudes and populated coastlines further increase risk in East Asia and the East China Sea [7, 8]. Consequently, improved prediction of TC track, intensity, wind radii, and upper-ocean response is an urgent requirement for disaster risk reduction. Track forecasting has improved substantially through better numerical weather prediction, satellite data assimilation, ensemble design, and, more recently, data-driven global weather models. Intensity forecasting has advanced more slowly [9, 10]. A central reason is that intensity is controlled not only by the large-scale atmospheric environment but also by air–sea enthalpy and momentum exchange, upper-ocean heat content, barrier-layer thickness, cold wakes, mesoscale eddies, and TC-induced currents. Strong winds mix the upper ocean, generate near-inertial currents, cool the sea surface, deepen the mixed layer, and modify vertical thermal and salinity structure; these responses feed back on surface fluxes and subsequent intensification [11, 12]. Satellite observations provide broad spatial coverage, but clouds and rain attenuate microwave signals, scatterometers saturate at high winds, and spaceborne products cannot directly resolve the subsurface ocean. The near-core wind and upper-ocean state therefore remain insufficiently constrained in many forecast initial conditions. Another challenge is the mismatch between the scales of the atmospheric and oceanic observing systems. The TC core and eyewall evolve over tens of kilometers and hours, whereas global atmospheric reanalyses and operational analyses are commonly available at quarter-degree and hourly resolution. Upper-ocean models provide finer horizontal grids, but assimilated observations are still spatially sparse. Even when temperature and salinity profiles are available, their forecast impact depends on where and when they are collected relative to the storm, the pre-existing barrier layer, and the ocean eddy field. Observations that are late, far from the sensitive region, or redundant with neighboring measurements may have little influence on the

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analysis. Adaptive observation is therefore not simply a coverage problem; it is a sequential decision problem involving motion, sampling time, energy, and coordination.

Current TC observing systems can be grouped into remote sensing, airborne, anchored, drifting, and autonomous underwater platforms [13, 14, 15]. Satellites are indispensable for monitoring storm position, cloud structure, precipitation, sea surface temperature, and large-scale winds [13]. They offer repeat coverage and synoptic context but cannot simultaneously resolve the surface wind maximum, the wave field, and the three-dimensional upper ocean beneath the core. Manned aircraft and dropwindsondes provide high-quality boundary-layer, eyewall, and upper-ocean information, but they are expensive, weather dependent, and limited to relatively short campaigns [14]. Small uncrewed aircraft can sample the turbulent eyewall boundary layer, but remain short-duration and range-limited [15]. Fixed moorings and buoys provide accurate time series but cannot follow a moving storm. Surface drifters sample Lagrangian pathways but provide little control over where measurements are made. Underwater gliders can remain at sea for weeks and have demonstrated value in TC research and assimilation, but their buoyancy-driven speed is much lower than storm translation speed [16, 17]. Wave gliders and conventional autonomous surface vehicles can communicate more easily, but they do not directly sample the subsurface current and stratification. The observational value of underwater platforms has nevertheless been demonstrated. Assimilation of glider profiles can improve the upper-ocean initial state and, in some cases, TC intensity forecasts [18, 19]. Strongly coupled ocean–atmosphere assimilation allows ocean observations to influence atmospheric analyses through cross-domain error covariances [20]. Studies using hypothetical current-observing networks and Marine JEDI show that current observations can alter coupled states and TC forecasts [21, 22]. The benefit is highly sensitive to the observing location and background ocean structure [23, 24, 25]. This motivates platforms that can move rapidly toward the predicted TC, adapt their sampling paths in response to local measurements, and switch between surface and subsurface operation.

A submersible unmanned vessel (SUV) combines characteristics of an unmanned surface vehicle and a submersible. It can use conventional propulsion for rapid surface transit, then lower the main hull below the free surface for improved seakeeping, sensor stability, and subsurface sampling [26, 27, 28, 29]. The Blue Whale, developed by the Second Institute of Oceanography, Ministry of Natural Resources, China, is an engineering reference for this class of platform [26]. It uses surface propulsion, four vector thrusters for low-speed subsurface maneuvering, and ballast, anchoring, and center-of-gravity control for submerged station keeping. Its public specifications include a maximum surface speed of about 23 kn (11.83 m/s), a submerged speed of about 3 kn (1.54 m/s) for at least 4 h, submerged standby for at least 72 h, a reference depth of about 60 m, and a minimum payload of about 800 kg. It can carry current profilers, CTD and biogeochemical sensors, and a rocketsonde for atmospheric profiles. This study does not attempt a digital twin of the current Blue Whale. Instead, the current platform provides the engineering baseline, and a future Blue Whale-class parameterization extends surface speed, endurance, depth, communication, and payload within explicitly identifiable scenarios. This distinction is important: an improved propulsion or energy system may be studied without attributing unverified performance to the present vehicle. The platform model represents speed, endurance, sampling depth, communication, energy, and safety constraints, while deliberately omitting high-fidelity hull hydrodynamics, wave loads, and actuator response.

The proposed observation cycle uses two coupled operating modes. In surface mode, the fleet moves rapidly toward the TC and enters the high-wind region. In subsurface mode, the vehicles dive to sampled depths and observe the TC-conditioned current field. The measured variables are therefore explicitly separated: at the sea surface, the SUVs observe wind speed and direction at 10 m above the sea surface; below the sea surface, they observe the horizontal current velocity components and the derived current speed at depths from 0 to 60 m. Figure 1 illustrates this transition. Four vehicles are used as the visual example, with equal angular spacing; the subsequent experiments extend the same parameter-sharing policy to 8, 12, and 16 vehicles. The stages share the platform and event data but are evaluated as separate tasks: the surface endpoint defines the operational handoff to the underwater rollout, while the dynamic dive, ballast, and resurfacing processes are outside the present motion model.

Classical graph-search, sampling-based, potential-field, and model predictive control methods have been used extensively for autonomous vehicle planning. Dijkstra and A* obtain optimal paths on known cost graphs, A* uses a heuristic to reduce search effort, and D* Lite repairs a previous solution when edge costs change [30, 31]. RRT and RRT* handle high-dimensional and nonholonomic constraints and approach feasible or asymptotically optimal solutions through sampling [32]. Artificial potential fields are computationally efficient but can become trapped in local minima [33]. Model predictive control explicitly handles dynamics, energy, and obstacle constraints over a receding horizon, but requires a reliable prediction model and repeated online optimization [34]. A comprehensive review of AUV planning is given by Panda et al. [35]. These methods remain valuable when the cost map and dynamics are known. In a TC environment, however, the wind and current fields vary continuously, vehicle ground velocity is the superposition of propulsion and ambient flow, and the moving cost graph would need frequent reconstruction. Each vehicle observes only a local patch, while multiple vehicles are coupled through coverage, redundancy, communication, and coastal safety. Traditional planning pipelines often decouple observation from path generation: a field estimate, cost map, or waypoint sequence is first constructed and then executed. In the RL formulation used here, the current measurement directly enters

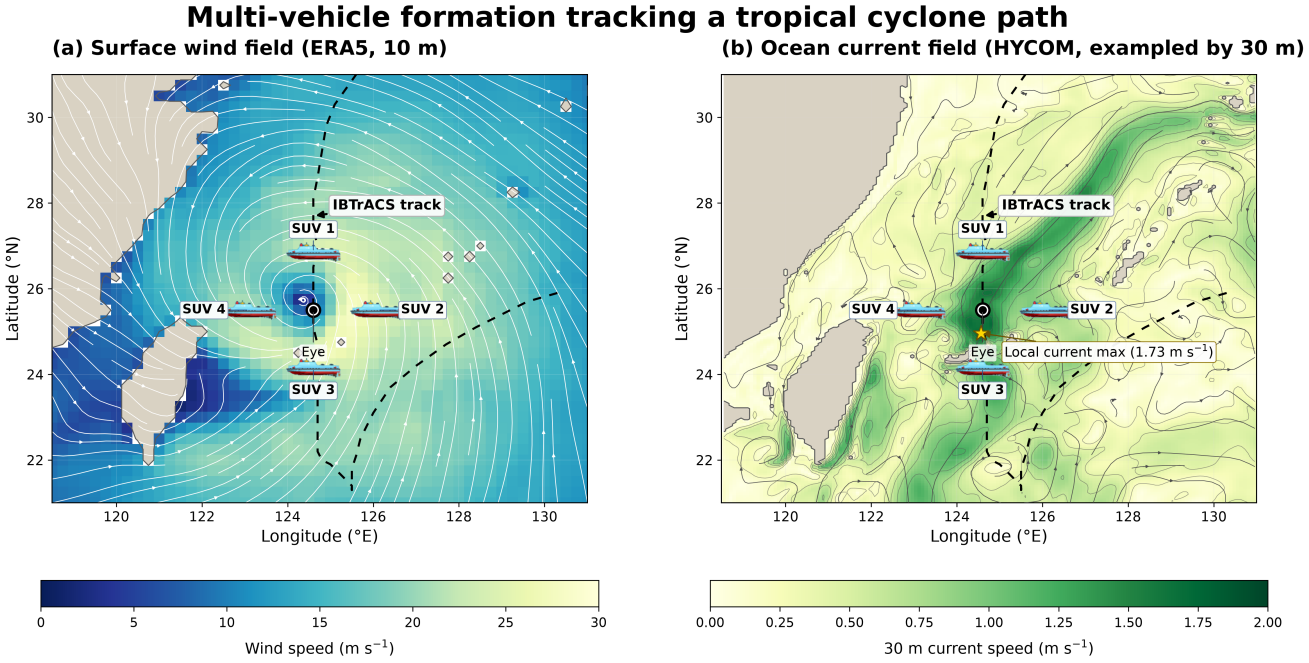


Figure 1: Coordinated surface–subsurface TC observation for a representative HINNAMNOR (2022) snapshot. The left panel shows four SUVs tracking the IBTrACS best track while sampling ERA5 winds at 10 m above the sea surface. The right panel shows the same formation sampling HYCOM currents at 30 m below the sea surface; this depth is shown as an illustrative example, whereas the complete experiments cover the seven depths from 0 to 60 m. The four-vehicle configuration is illustrative, with equal angular spacing; subsequent experiments evaluate fleets of 8, 12, and 16 vehicles.

the policy state and changes the next action; that action changes the vehicle position and produces a new measurement. Observation and motion planning are therefore coupled in one feedback law rather than treated as separate sequential modules. Reinforcement learning provides an alternative for closed-loop decision making under uncertainty. A policy is optimized through repeated interaction with the environment and can approximate a state-to-action mapping without explicitly solving the full stochastic transition model. RL has achieved strong results in language-model alignment and reasoning [36, 37, 38], legged locomotion and dexterous manipulation [39, 40], drone racing [41], and high-dimensional games [42, 43]. Independent PPO has been competitive in cooperative multi-agent benchmarks and games [44, 45], including the StarCraft Multi-Agent Challenge [46]. In the ocean domain, multi-AUV adaptive sampling and cooperative control are established applications [47]; RL has more recently been applied to underwater target tracking and cooperative AUV tasks [48, 49, 50]. Hybrid adaptive methods also remain useful for multi-AUV cooperative hunting in obstacle-rich environments [51].

Three unresolved gaps motivate this study. First, previous planning studies often reconstruct the ocean response with simplified analytical or reduced-order flow models [52, 53] and therefore do not explicitly use event-specific background eddies, fronts, vertical shear, or non-stationary current structures. Second, TC track forecasts retain finite position and timing errors [23, 24, 25]; fixed pre-deployment and waiting strategies can therefore miss the actual high-wind region. A practical system must allow the fleet to move rapidly from its deployment location to the high-wind region and then descend for subsurface sampling. Third, the closest previous multi-AUV active-observation study used four vehicles [53]; fleet-size sensitivity therefore remains insufficiently explored, limiting the number of simultaneous and spatially distributed observations. The present study therefore evaluates four fleet sizes: 4, 8, 12, and 16 SUVs.

From a control perspective, the coupled wind–current environment is nonstationary, spatially heterogeneous, and stochastic. Exact online solution of the full multi-agent transition model is therefore impractical under partial observability and limited computation. We formulate the decision problem as stochastic-approximation-based policy optimization, in which neural networks approximate the policy and value functions. IPPO learns the closed-loop policy from repeated interaction with event-specific environmental fields [54, 44, 45]. Three earlier results define the starting point of this work. Qi et al. developed a forecast-path correction model that used AUV current observations to infer TC information and update a tracking path [52]. A subsequent multi-AUV active observation framework included forecast tracks, background currents, travel time, energy, and redundancy, but represented the ocean with an analytical model [53]. These studies provide the center-inference and task-planning foundations for the present work; the event-specific surface–subsurface loop developed here addresses the gaps identified above.

2. Problem Formulation

2.1. Observation Task and Design Requirements

The objective is to coordinate a fleet of submersible unmanned vessels (SUVs) to observe a moving tropical cyclone (TC) across the sea surface and upper ocean. Each SUV first navigates on the surface toward the predicted TC center and high-wind region. After entering a predefined high-wind observation window, the fleet descends to a specified depth and samples the TC-induced current field. The framework is designed to satisfy four operational requirements:

1. rapid correction of initial deployment errors using surface propulsion;
2. decentralized tracking of the moving TC using onboard wind observations and inter-vehicle observation sharing;
3. subsurface observation at ocean depths of 0, 10, 20, 30, 40, 50, and 60 m below the sea surface;
4. safe operation under propulsion-energy, communication, action-smoothness, and coastline constraints.

The surface maximum speed is varied between 12 and 20 m/s, while the submerged maximum propulsion speed is 1.5 or 2.0 m/s. This difference creates an asymmetric decision problem: active propulsion is necessary at the surface, whereas underwater observation must also consider passive advection by the ambient current.

2.2. Problem Definition

The fleet contains N SUVs, where $N \in \{4, 8, 12, 16\}$, indexed by $i \in \{1, 2, \dots, N\}$. The position, propulsion velocity, and operating mode of SUV i at time t are denoted by $\mathbf{x}_i(t)$, $\mathbf{p}_i(t)$, and $m_i(t) \in \{\mathrm{s}, \mathrm{u}\}$, respectively, where s and u denote the surface and subsurface modes. The propulsion speed satisfies

$$\|\mathbf{p}_i(t)\| \le \begin{cases} V^{\mathrm{s}}_{\max}, & m_i(t) = \mathrm{s}, \\ V^{\mathrm{u}}_{\max}, & m_i(t) = \mathrm{u}. \end{cases} \tag{1}$$

The horizontal motion of the SUV is modeled as

$$\dot{\mathbf{x}}_i(t) = \begin{cases} \mathbf{p}_i(t), & m_i(t) = \mathrm{s}, \\ \mathbf{p}_i(t) + \mathbf{u}_{\mathrm{ocean}}\left(\mathbf{x}_i(t), z_i, t\right), & m_i(t) = \mathrm{u}, \end{cases} \tag{2}$$

where $z_i \in \{0, 10, 20, 30, 40, 50, 60\}$ m and $\mathbf{u}_{\mathrm{ocean}}(\cdot)$ is the HYCOM current at the corresponding position and depth. Thus, the underwater ground velocity is the superposition of active propulsion and background advection.

For numerical implementation, the continuous model is integrated explicitly at the decision interval $\Delta t = 10$ min:

$$\mathbf{x}_{i,k+1} = \begin{cases} \mathbf{x}_{i,k} + \mathbf{p}_{i,k}\Delta t, & m_{i,k} = \mathrm{s}, \\ \mathbf{x}_{i,k} + \left[\mathbf{p}_{i,k} + \mathbf{u}_{\mathrm{ocean}}(\mathbf{x}_{i,k}, z_i, t_k)\right]\Delta t, & m_{i,k} = \mathrm{u}. \end{cases} \tag{3}$$

The current field is evaluated at the beginning of each interval and interpolated to the current vehicle position. For the active underwater reward, we write this pre-move current as $\mathbf{u}_{i,k} = \mathbf{u}_{\mathrm{ocean}}(\mathbf{x}_{i,k}, z_i, t_k)$.

This equation is deliberately a reduced-order horizontal point-mass kinematic–advection model rather than a six-degree-of-freedom rigid-body dynamic model. It does not resolve added mass, hydrodynamic drag, roll, pitch, yaw or heading dynamics, buoyancy and ballast dynamics, thruster transients, wave loads, or actuator delay. This simplification is consistent with the information content of the environmental forcing. ERA5 winds are available at 0.25° and hourly resolution, whereas HYCOM currents are available at approximately 0.04° by 0.08° and three-hourly resolution. These fields are interpolated to the 10 min decision interval and to each vehicle position, but such interpolation does not recover high-frequency wave forcing, local wake effects, or unobserved vehicle attitude and actuator states. Introducing a detailed six-degree-of-freedom model would therefore require platform-specific hydrodynamic coefficients and forcing information that are not commensurate with the present data, and could imply a level of numerical precision unsupported by the inputs. The reduced model retains the first-order effects relevant to mission-level planning: mode-dependent speed limits, propulsion commands, ambient-current advection in subsurface mode, and coastal constraints. Accordingly, the results are interpreted as mission-planning and coordination evidence rather than as predictions of attitude, structural loading, or low-level controller response. Coupling this planning layer to an experimentally calibrated six-degree-of-freedom vehicle model is left for future work.

The local observation of SUV i is

$$\mathbf{o}_i(t) = \left[\mathbf{s}_i(t), \mathbf{v}_i(t), \mathbf{h}_i(t)\right], \tag{4}$$

where $\mathbf{s}_i(t)$ contains the observation used in the corresponding stage and $\mathbf{v}_i(t)$ is the normalized commanded propulsion velocity, $\mathbf{v}_i = \mathbf{p}_i / V^{m_i}_{\max}$; it is not the total ground velocity. In the final surface experiments, $\mathbf{s}_i(t)$ consists only of the normalized wind patches at 10 m above the sea surface sampled by the N SUVs, and no historical frame is used. Surface operation assumes idealized, lossless, and instantaneous inter-vehicle communication: every SUV receives the wind observations acquired by all other SUVs in the fleet as part of its policy input. In the underwater active-control comparison, no inter-vehicle communication is assumed. Accordingly, $\mathbf{s}_i(t)$ consists only of SUV i's local current speed and components at its assigned ocean depth, while $\mathbf{h}_i(t)$ contains a five-frame history of that same vehicle's local current. Each underwater SUV therefore knows neither the current observations nor the positions or actions of the other SUVs. The complete reanalysis field, the global best-track center, and future environmental states are not provided to the policy. Accordingly, the surface observation dimension is $d^{\mathrm{s}}_o = 3N + 2$. The underwater active controller uses a 1×1 local-current observation plus a five-frame history and two normalized propulsion components, giving $d^{\mathrm{u}}_o = 5 \times 3 + 2 = 17$.

During training, the reward function can access the distance between each SUV and the best-track TC center. This distance supervises the association between local 10 m wind patterns and proximity to the TC center/high-wind region; the center coordinate itself is not an actor input. During validation, rewards do not influence actions. The frozen policy receives only the shared fleet-level wind samples and must track an unseen TC, thereby testing whether the learned wind-to-center relation generalizes.

The normalized action is

$$\mathbf{a}_i(t) \in [-1, 1]^2, \tag{5}$$

and is projected onto the unit disk before being mapped to propulsion in both modes:

$$\tilde{\mathbf{a}}_i(t) = \frac{\mathbf{a}_i(t)}{\max\left(1, \|\mathbf{a}_i(t)\|_2\right)}, \qquad \mathbf{p}_i(t) = V_{\max}\tilde{\mathbf{a}}_i(t). \tag{6}$$

This mapping ensures that the total propulsion speed never exceeds the mode-specific speed limit.

The cooperative task is formulated as a decentralized partially observable Markov decision process (Dec-POMDP):

$$\mathcal{M} = \langle \mathcal{I}, \mathcal{S}, \{\mathcal{O}_i\}, \{\mathcal{A}_i\}, P, R, \gamma \rangle, \tag{7}$$

where $\mathcal{I}$ is the set of SUVs, $\mathcal{S}$ is the joint environment state, $\mathcal{O}_i$ and $\mathcal{A}_i$ are the local observation and action spaces of vehicle i, P is the transition probability, r_i is the local reward contribution, $R = (1/N)\sum_{i=1}^{N} r_i$ is the shared team reward, and γ is the discount factor.

2.3. Observation Objectives and Operational Conditions

The framework is designed around several observation requirements: surface TC tracking, local-current sampling, spatial coverage, information quality, energy efficiency, redundancy reduction, and coastal safety. These requirements are not optimized through one fixed-weight global objective. Instead, the surface and subsurface stages use the task-specific rewards defined in Section 4. Coverage, information quality, and redundancy are also reported as evaluation metrics. The final training objective is restricted to the stage-specific rewards defined below rather than a single weighted sum of all these quantities.

The propulsion bound in Eq. (1) is enforced explicitly, where $\mathbf{p}_i(t)$ is the propulsion velocity rather than the total ground velocity.

Energy use is reported through an E_3 propulsion proxy. In active underwater training it also enters the reward as a soft penalty; the surface policy has no hard terminal energy budget and uses action smoothing instead of an explicit power term. Coastal safety is handled through offshore-only rejection sampling during deployment, a soft nearshore penalty, and an action-level safety filter that removes the onshore component within the specified safety margin. The main surface experiments assume ideal sharing of the wind patches sampled by all SUVs. The final subsurface comparison includes active IPPO, random control, and passive drift. Passive drift provides the no-propulsion reference for evaluating whether active control improves the observation objective.

3. Data and Simulation Environment

3.1. Reanalysis and Best-Track Products

ERA5 supplies hourly zonal and meridional wind components at 10 m above the sea surface on a regular 0.25° grid [1] [55]. Wind speed and direction are calculated from u_{10} and v_{10}. HYCOM supplies three-hourly zonal and meridional currents at ocean depths of 0, 10, 20, 30, 40, 50, and 60 m below the sea surface [2] [56]. The native horizontal grid is approximately 0.04° in latitude by 0.08° in longitude. IBTrACS provides TC time, center position, maximum wind, minimum pressure, and related metadata at mostly 3 or 6 h intervals [3] [57]. The common study domain is 98.7°E–171.6°E and 6.8°N–56.7°N. The overall event set spans 19 November 2007 to 3 December 2025 UTC. Thus, the surface observation variable is the 10 m wind speed and direction, whereas the subsurface observation variables are the horizontal current velocity, represented by `water_u` and `water_v`, and the derived current speed over the 0–60 m layer. Table 1 summarizes the products, variables, native grids, and temporal resolutions used to construct the event-specific environments.

3.2. Spatiotemporal Alignment and Track Segmentation

All fields are mapped to a common simulation clock. Surface atmospheric fields are linearly interpolated from hourly values, and ocean currents are linearly interpolated from three-hourly values to the 10 min decision interval. This is an offline reanalysis simulation: adjacent reanalysis samples are used to reconstruct the field at the simulated time, and the policy receives only the interpolated local value at that time, not a future time index or full future field. This construction does not emulate the data latency of a real-time forecast system. At each decision time, the observation operator uses the reconstructed field, while the best-track center is used only for training supervision and evaluation. Any NaN or infinity produced by interpolation or coastal masking is replaced by a pre-specified neutral value after finite-difference gradients are computed. The resulting observations are checked for finiteness before entering the policy.

Although the native grids and sampling intervals are much coarser than the 10 min decision step, the environment performs online spatial and temporal interpolation at every

[1] ERA5 download portal: https://cds.climate.copernicus.eu/datasets/reanalysis-era5-single-levels.

[2] HYCOM GLBy0.08 archive: https://www.hycom.org/data/glby0p08/expt-93pt0; HYCOM ESPC-D-V02 archive: https://www.hycom.org/data/espcd-v02.

[3] IBTrACS download portal: https://www.ncei.noaa.gov/products/international-best-track-archive.

Table 1
Data used to construct the TC observation environment. Each event retains its own time window; the common time span is reported for the complete archive. The target archive combines ERA5 surface winds, HYCOM currents at seven depths, and IBTrACS best tracks.

Data product	Source	Variables used	Native grid	Native time
Surface wind at 10 m above sea level	ERA5	u_{10}, v_{10}, speed, direction	$0.25° \times 0.25°$	Hourly
Current at 0 m ocean depth	HYCOM	water_u, water_v, speed	$\sim 0.04°$ lat $\times 0.08°$ lon	3-hourly
Current at 10 m ocean depth	HYCOM	water_u, water_v, speed	$\sim 0.04°$ lat $\times 0.08°$ lon	3-hourly
Current at 20 m ocean depth	HYCOM	water_u, water_v, speed	$\sim 0.04°$ lat $\times 0.08°$ lon	3-hourly
Current at 30 m ocean depth	HYCOM	water_u, water_v, speed	$\sim 0.04°$ lat $\times 0.08°$ lon	3-hourly
Current at 40 m ocean depth	HYCOM	water_u, water_v, speed	$\sim 0.04°$ lat $\times 0.08°$ lon	3-hourly
Current at 50 m ocean depth	HYCOM	water_u, water_v, speed	$\sim 0.04°$ lat $\times 0.08°$ lon	3-hourly
Current at 60 m ocean depth	HYCOM	water_u, water_v, speed	$\sim 0.04°$ lat $\times 0.08°$ lon	3-hourly
Best track	IBTrACS	Position, time, wind, pressure	Point track	Mainly 3/6-hourly

Note: the final subsurface matrix uses paired events for which the selected depth is non-empty. The main policy uses ERA5 winds and HYCOM currents only; temperature and salinity are not policy inputs.

decision time. Each SUV queries the interpolated field at its current position and at the corresponding simulation time, and the same interpolated values are used for its observation and for current-induced motion. These values are treated as the best available estimates from the archive under the stated interpolation scheme, rather than as resolved high-resolution truth or statistically optimal reconstructions of unobserved sub-grid variability.

The 10 min decision interval is selected as a compromise between control responsiveness and the information content of the source products. A longer interval would reduce the frequency of trajectory correction and degrade decision precision. A substantially shorter interval would increase decision and execution overhead and would substantially increase the number of decisions per episode, the required interaction samples, and the training workload. It would also require temporal detail beyond what can be supported by interpolating hourly ERA5 winds and three-hourly HYCOM currents. Using such an interval would therefore create a false impression of higher temporal resolution without introducing genuinely new environmental information.

Each IBTrACS track is first linearly interpolated to the 10 min simulation interval. The cumulative along-track distance is computed using the local equirectangular approximation

$$\Delta s_k = \sqrt{\left[111\Delta\lambda_k \cos(\varphi_k)\right]^2 + \left(111\Delta\varphi_k\right)^2}, \qquad (8)$$

where $\Delta\lambda_k$ and $\Delta\varphi_k$ are the longitudinal and latitudinal increments in degrees. The constant 111 km is the mean meridional arc length per degree, and $\cos(\varphi_k)$ converts a longitudinal degree to the corresponding zonal distance at latitude φ_k. The target length of each segment is sampled uniformly from 300–500 km. The start of the next segment is shifted by a fixed 250 km along the track, producing an overlap of 50–250 km and an average overlap of approximately 150 km. A new segment is initialized only when at least 300 km of track remains; the final residual is discarded. Because segment endpoints are assigned to the nearest 10 min interpolation point, the realized segment lengths are 292.1–507.1 km (median 396.4 km).

The 500-TC archive yields 7,248 segments from 495 TCs with sufficient track length. The training split contains 450 TCs, of which 445 contribute at least one usable segment, yielding 6,480 training segments. The held-out test split contains 50 TCs and 768 segments. The training and test sets are separated at the TC level, so segments from the same TC never appear in both sets. Segment durations range from 2.2 to 103 h, with median and mean durations of 16.5 and 20.0 h, respectively.

Track segmentation is not merely a data-processing choice; it reflects the physical operating limit of the surface platform. The surface endurance of an SUV is finite, so one deployment cannot observe the complete life cycle of a TC. A 300–500 km along-track segment instead represents a self-contained mission in which the fleet can correct forecast-induced displacement, reach a physically relevant part of the TC, and collect a sustained observation sequence before the surface energy budget is exhausted. After entering the near-core region, the vehicles can dive and sample the upper ocean at several depths. The Blue Whale-class payload can accommodate current profilers, CTD, and biogeochemical sensors [26], so the complete observing concept supports temperature, salinity, current velocity, nutrients, oxygen, and other biogeochemical variables. These measurements can characterize both the physical and biogeochemical responses of the upper ocean to TC forcing. The control experiments deliberately use current velocity only as the decision-making observation, avoiding model dependence on variables that are not consistently available across the complete 500-TC archive; this restriction does not imply that the platform is limited to current measurements in a real deployment.

We adopt an 800 km training radius as an explicit upper-bound assumption on the position error of the operational TC track forecast used to guide deployment. It represents a scenario in which forecast uncertainty may place the intended observation region as much as 800 km away from the deployed fleet. During training, both radial distance and

bearing are sampled as random points within this disk. The 100, 200, 400, 600, and 800 km validation radii then provide a controlled sequence from mild to severe forecast-induced deployment error: 100 km approximates an accurate forecast, whereas 800 km corresponds to the assumed upper bound. Applying the same trained policy at all five radii, without retraining, quantifies how tracking and observation performance degrade as the deployment error becomes more severe. We do not train a separate model for the 200 km validation condition. The 0–800 km randomized training distribution already contains the 200 km deployment error, and the same trained policy is subsequently evaluated at 200 km without retraining. This design reduces dependence on one fixed initial displacement and provides a stronger test of generalization across different realizations of forecast-induced deployment error.

This 800 km range applies to the surface approach stage and is not used as the subsurface deployment radius. Operationally, underwater observation begins only after surface tracking has brought the fleet into the TC core region. Accordingly, the subsurface experiments initialize the four vehicles within 200 km of the TC center. This two-stage design separates long-range correction of track-forecast error on the surface from near-core current observation underwater.

The 450/50 division is also defined at the TC level and is used to estimate generalization to previously unseen TCs. No segment or environmental field from a held-out TC is exposed during training. The 50 unseen TCs therefore emulate the practical deployment condition in which the policy encounters an unknown future TC, while the radius series separately characterizes sensitivity to the magnitude of forecast-induced deployment error.

4. Multi-Agent Reinforcement Learning Method

4.1. Blue Whale-Class Platform Modeling

The platform state includes horizontal position, propulsion velocity, local surface or subsurface current, accumulated path length, accumulated propulsion-energy proxy, and, when enabled, distance to the coastline. The surface maximum speed is varied between the original 20 m/s research assumption and a 12 m/s scenario close to the current Blue Whale's public 23 kn. Subsurface maximum propulsion speed is 1.5 or 2.0 m/s. The latter is a clearly marked future-upgrade case. State propagation follows the mode-dependent speed bound and horizontal motion model in Eqs. (1) and (2), with actions mapped to propulsion through Eq. (6). Surface motion neglects the direct current contribution because the 12–20 m/s propulsion speed is much larger than the typical surface-current contribution, and because the surface stage is intended to represent TC approach and high-wind-region tracking. In contrast, the subsurface model explicitly retains ambient advection because the 1.5–2.0 m/s propulsion speed is of the same order as the TC-induced current. The reduced point-mass formulation and its limitations are stated after Eq. (2). Table 2 separates the public engineering reference from the two simulation scenarios and identifies which values are modeling assumptions rather than certified platform specifications.

4.2. Decentralized Partially Observable Decision Process

The cooperative observation task uses the Dec-POMDP defined in Eq. (7).

For each vehicle i, the observation contains:

1. the normalized wind patches at 10 m above the sea surface sampled by all SUVs in the main surface experiments, or its own local current observation at the assigned ocean depth in the active underwater comparison;
2. its own velocity in the surface configuration;
3. a five-frame history of local current speed and components when the underwater active controller is used.

The final wind-only surface configuration is memoryless and assumes ideal communication, so every SUV receives the $3N$ wind-observation values acquired by the full fleet plus its own two normalized velocity components. By contrast, the underwater controller receives only its own local current and velocity channels; the shared policy parameters do not imply that observations, states, or actions are shared among vehicles. The main experiment does not provide the complete wind or current field, the global best-track center, or future reanalysis frames to the actor. The best track is used only to define the training target and to compute evaluation metrics.

The policy operates as a closed perception–decision–action loop rather than following a precomputed static route. At every 10 min decision interval, it observes the current environment, generates a new action, moves the vehicle, observes the updated environment, and repeats:

$$\begin{aligned}\mathbf{o}_{i,t} &\longrightarrow \mathbf{a}_{i,t} = \pi_\theta(\mathbf{o}_{i,t}) \\ &\longrightarrow \mathbf{x}_{i,t+1} \longrightarrow \mathbf{o}_{i,t+1}.\end{aligned} \tag{9}$$

This feedback process allows the fleet to revise its motion as the wind and current fields evolve. The observation is not only an output recorded after motion; it is also the next control input that determines subsequent motion. Here, "supervision" refers to this measurement-driven feedback role rather than to supervised learning with target action labels: the policy is trained from scalar rewards, but each new observation directly supervises the next decision through the closed-loop state. Because the coupled environment is nonstationary and stochastic, this observation–planning coupling is the principal distinction from static or decoupled trajectory-planning pipelines. The same mechanism supports adaptation to changing local observations, partial observability, and deployment uncertainty.

4.3. Surface and Subsurface Reward Design

The surface task is to reduce the distance between the SUV fleet and the TC center while penalizing excessive

Table 2
Public Blue Whale reference and two simulation scenarios. The two simulation columns are modeling assumptions, not certified vehicle data.

Property	Public reference	High speed scenario	Current speed scenario
Role	Engineering reference	Future SUV	Near-current SUV
Surface speed	23 kn, ≈11.83 m/s	20 m/s	12 m/s
Submerged speed	3 kn, ≈1.54 m/s	1.5 or 2.0 m/s	1.5 or 2.0 m/s
Submerged endurance	≥4 h at 3 kn; ≥72 h standby	20 h scenario	20 h scenario
Sampling depth	≈60 m	0–60 m (7-layer)	0–60 m (7-layer)
Payload	≥800 kg	800 kg	800 kg
Fleet size	Single platform	4/8/12/16	4/8/12/16
Decision interval	Not specified	10 min	10 min
Formation	–	100 km ring, $360°/N$	100 km ring, $360°/N$

separation from the target region. Let d_i denote the distance from SUV i to the TC center. The surface reward used in the implementation is

$$\begin{aligned} r_i^{\mathrm{surf}} = & -\frac{\min(d_i, 300)}{100} + b_{\mathrm{near}}\mathbf{1}_{d_i<100} \\ & -\frac{W_{\mathrm{form}}\max(0, d_i - R_{\mathrm{form}})}{100} \\ & -\lambda_u\|\mathbf{a}_{i,t} - \mathbf{a}_{i,t-1}\|_2^2 + r_i^{\mathrm{coast}}, \end{aligned} \tag{10}$$

where the first term is a bounded proximity cost and the second term provides a discrete near-center bonus with $b_{\mathrm{near}} = 1$. The third term adds an additional radius-excess penalty beyond $R_{\mathrm{form}} = 100$ km, and the fourth term penalizes abrupt action changes with $\lambda_u = 0.05$. The term r_i^{coast} represents the coastal-safety reward defined below. The 300 km cap prevents distant errors from producing an excessively steep gradient and reduces premature policy collapse during early training. After aggregation over SUVs, the team average reward is clipped to $[-5, 2]$.

For subsurface observation, the objective is not to follow the surface TC center directly. Low-speed SUVs cannot generally overtake a translating TC, and the vertical projection of the surface center is not necessarily the location of the strongest subsurface current. For the active-control comparison, the current-seeking reward is

$$\begin{aligned} r_i^{\mathrm{sub}} = & \min\left(\frac{\|\mathbf{u}_{i,k}\|}{V_{\mathrm{norm}}}, R_{\max}\right) \\ & -\lambda_E\|\tilde{\mathbf{a}}_i\|_2^2 + \beta_{\parallel}\max(0, \cos\theta_i) \\ & + r_i^{\mathrm{coast}}, \end{aligned} \tag{11}$$

where $V_{\mathrm{norm}} = 1.0$ m/s, $R_{\max} = 2.0$, $\lambda_E = 0.05$, and $\beta_{\parallel} = 0.5$. The angle θ_i denotes the direction difference between propulsion and the pre-move current $\mathbf{u}_{i,k}$, so the positive-alignment term is activated only when the SUV pushes downstream; formally, $\cos\theta_i = (\tilde{\mathbf{a}}_i \cdot \mathbf{u}_{i,k})/(\|\tilde{\mathbf{a}}_i\|_2\|\mathbf{u}_{i,k}\|_2)$ when the denominator is nonzero. The current-intensity reward is therefore evaluated at $\mathbf{x}_{i,k}$ before the state update. For the local maximum-current objective, passive drift is retained as the no-propulsion reference because it follows the ambient flow directly and does not depend on a learned reward policy.

4.4. Coastal Safety and Action Smoothing

The land mask is converted into a nonnegative offshore-distance field. A vehicle inside land receives a large penalty, and a vehicle within the buffer distance D_{buf} receives

$$r_i^{\mathrm{coast}} = \begin{cases} -P_{\mathrm{land}}, & D_i = 0, \\ -W_b\max(0, 1 - D_i/D_{\mathrm{buf}}), & 0 < D_i < D_{\mathrm{buf}}, \\ 0, & D_i \geq D_{\mathrm{buf}}, \end{cases} \tag{12}$$

where D_i is the offshore distance and P_{land} and W_b are positive penalty coefficients.

Three complementary mechanisms are used to reduce coastline intrusion: offshore-only rejection sampling during deployment, a soft nearshore penalty, and an action-level safety filter. The current main experiments use a minimum deployment distance of 100 km, a 100 km nearshore buffer, and a 100 km safety margin. The deployment step removes invalid initial states, the reward discourages trajectories near land, and the safety filter removes the onshore action component when an SUV approaches the shoreline. These mechanisms reduce land exposure but are not posed as an unconditional hard constraint on every vehicle position.

To reduce oscillatory control, the main surface experiments use an exponential action filter:

$$\mathbf{a}_t^{\mathrm{filt}} = (1 - \beta)\mathbf{a}_t^{\mathrm{raw}} + \beta\mathbf{a}_{t-1}^{\mathrm{filt}}, \tag{13}$$

where $\beta \in [0, 1)$ controls the degree of temporal smoothing. The filter uses only the current and previous actions and therefore does not introduce future-information leakage.

4.5. Parameter-Sharing Independent PPO

All SUVs contribute to the same observation and tracking objective, and the training return is defined as the team-average reward. The problem is therefore a fully cooperative task, rather than a competitive or mixed cooperative–competitive task. This cooperative reward structure motivates the use of IPPO, in which decentralized vehicle actions are optimized against a common team objective while policy parameters are shared across vehicles.

In this paper, IPPO denotes the cooperative and parameter-sharing implementation of independent PPO. All SUVs use the same actor and critic weights, while their observations and action samples remain vehicle-specific. The main surface model uses a memoryless multilayer perceptron. The active underwater controller uses a five-frame current history. The current implementation uses a shared cooperative return, obtained by averaging the per-SUV rewards, and optimizes a single clipped PPO objective over the factorized multi-agent action vector:

$$L^{\mathrm{CLIP}}(\theta) = \mathbb{E}_t\Big[\min\Big(\rho_t(\theta)\hat{A}_t, \; \mathrm{clip}\left(\rho_t(\theta), 1-\epsilon, 1+\epsilon\right)\hat{A}_t\Big)\Big], \tag{14}$$

where

$$\rho_t(\theta) = \prod_{i\in\mathcal{I}} \frac{\pi_\theta(a_{i,t} \mid o_{i,t})}{\pi_{\theta_{\mathrm{old}}}(a_{i,t} \mid o_{i,t})}, \tag{15}$$

and $\hat{A}_t$ is estimated by generalized advantage estimation from the team-average reward. The actor objective is complemented by a clipped value loss and an entropy bonus.

In the reported implementation, IPPO denotes a parameter-sharing, cooperative PPO variant: the actor and critic are shared across vehicles, actions are sampled independently for each vehicle, the joint action log-probability is used in the PPO ratio, and the team-average reward defines the shared return. The shared critic is evaluated separately for each local SUV observation, and the resulting per-agent values are summed to form the training value. No global-state centralized critic is used. During execution, each SUV computes its action from its own observation and the shared policy network. The main surface configuration assumes ideal information sharing among the SUVs. Passive drift uses zero propulsion and therefore has no learned actuator policy.

Off-policy SAC and TD3 were not used for the final multi-SUV comparison. Both require a replay buffer, and a multi-agent transition contains the joint observation and action of up to 16 vehicles. The memory footprint therefore scales with replay capacity, fleet size, and the number of parallel environments. The final experiments prioritize stable large-scale parallel on-policy sampling and consequently use parameter-sharing IPPO. This choice is an algorithm-design constraint rather than a claim that IPPO is universally superior to SAC or TD3 in sample efficiency.

The complete reward coefficients, safety parameters, network dimensions, and PPO hyperparameters are summarized in Appendix A. The project repository and software package sources are listed in the Code and package availability section.

4.6. Surface Evaluation, Generalization, and Safety

The center-distance reward is available only during training. It does not enter the actor observation and is absent during validation. Consequently, the surface results test a specific form of generalization: a frozen policy must infer where to move from shared wind observations in an unseen TC. The policy is not asked to read the best-track center, and it cannot exploit a reward computed from that center. This distinction is essential for interpreting the 100–800 km curves as forecast-error tolerance rather than as supervised center tracking.

Once the high-wind region is reached, diving provides the second part of the sampling objective. The subsurface current there is expected to contain the upper-ocean response generated by the TC forcing and its preceding passage, including near-inertial and mesoscale variability. Sampling that dynamically conditioned region is expected to be more informative than descending in an unrelated background ocean. The present study evaluates the observation task itself; it does not quantify the subsequent impact of these measurements in a data-assimilation or forecast system.

The final surface matrix uses only local wind observations and evaluates 4-, 8-, 12-, and 16-SUV formations. The vehicles in each formation have equal angular spacing of $360°/N$ on a 100 km ring. During training, the formation-center offset is sampled uniformly inside an 800 km disk around the initial TC center, representing the assumed upper bound of track-forecast position error. The trained policies are then evaluated without retraining at fixed formation-center offsets of 100, 200, 400, 600, and 800 km with random bearings. The coastal configuration combines offshore-only deployment, a soft nearshore penalty, and a 100 km action-level safety filter to reduce coastline intrusion while preserving TC-region tracking.

For each fleet-size, deployment-radius, and speed setting, the 768 held-out segments were first averaged within each training seed. Means and sample standard deviations were then computed across the ten independent seeds. Paired comparisons between 12 and 20 m/s used two-sided paired t-tests on the ten seed-level values. Because the twenty fleet-size/radius comparisons were tested separately without multiplicity correction, the resulting p-values are interpreted as exploratory evidence rather than confirmatory hypothesis tests.

5. Simulation and Results

5.1. Surface Active Tracking

Using the surface protocol in Section 4.6, we evaluate the extent to which the trained policies reduce forecast-induced deployment errors and approach the moving TC center and high-wind region.

All 12 and 20 m/s comparisons use the same 450 training TCs, 50 held-out TCs, and 768 test segments. The observation, network, reward, formation, filtering, and coastal-safety settings are held fixed, and only the speed limit is changed. Two complementary tracking criteria are evaluated: *time-averaged distance*, which measures whether the fleet remains near the moving TC throughout an episode, and *final distance*, which measures whether it reaches the TC center by the end of the response window. Paired tests compare 12 and 20 m/s at the same fleet size, deployment radius,

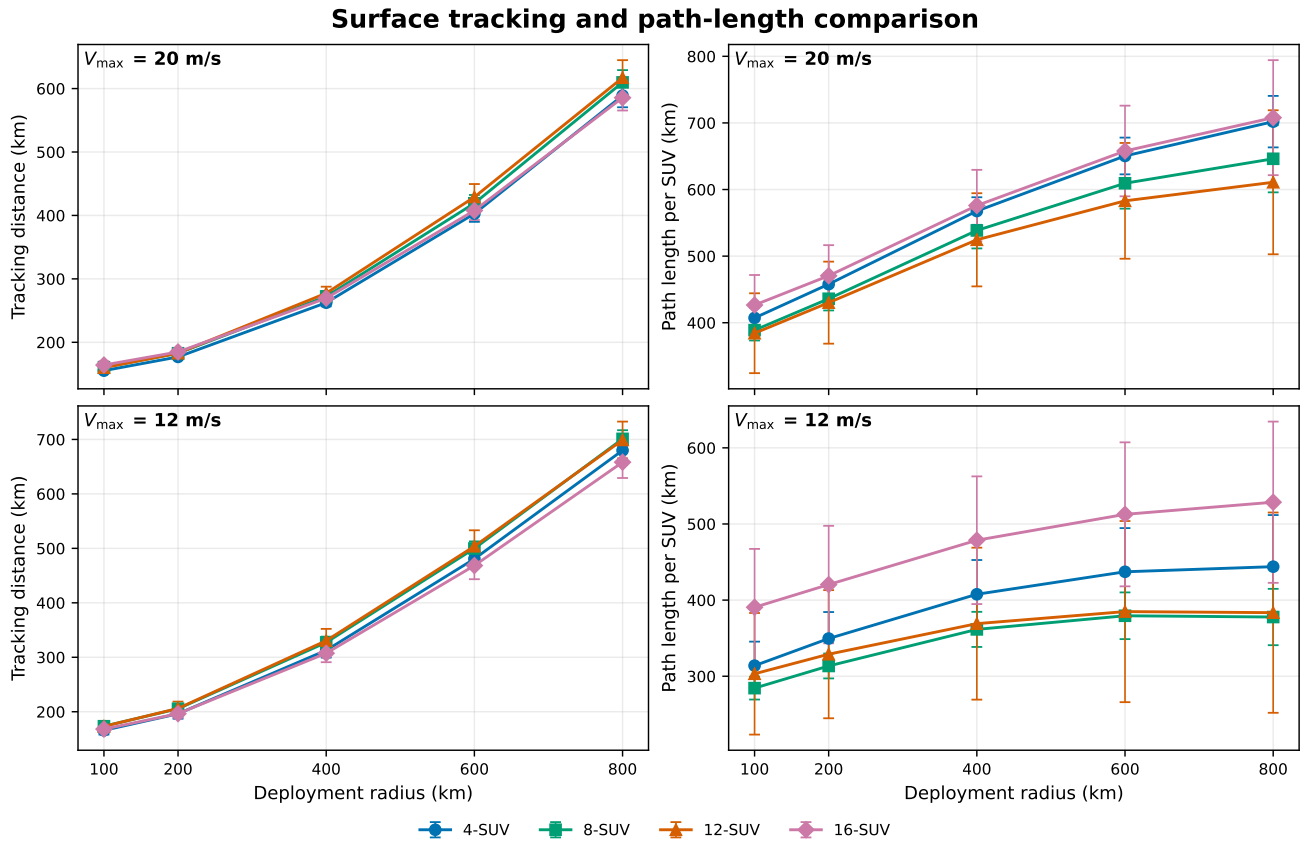


Figure 2: Interim surface tracking comparison. The final version will use the time-averaged and final TC-center distances shown in the planned two-panel layout.

and seed. Each reported value summarizes ten independent training runs (ten random seeds).

Across the twenty matched fleet-size/radius comparisons, the 20 m/s policy has the smaller mean time-averaged distance in all 20 cases. The median paired difference ($d_{12} - d_{20}$) is +51.8 km. Uncorrected two-sided paired tests indicate lower distance at 20 m/s in 19 cases and at 12 m/s in none of the cases at $p < 0.05$. Because these comparisons were not adjusted for multiplicity, they provide exploratory rather than confirmatory evidence.

Averaged over the four fleet sizes, the differences (12 m/s minus 20 m/s) at the five radii are: 100 km: $\Delta d = +10.1$ km, $\Delta L = -78.7$ km, $\Delta\bar{v} = -1.74$ m/s; 200 km: $\Delta d = +19.3$ km, $\Delta L = -95.5$ km, $\Delta\bar{v} = -2.07$ m/s; 400 km: $\Delta d = +49.1$ km, $\Delta L = -147.4$ km, $\Delta\bar{v} = -3.02$ m/s; 600 km: $\Delta d = +73.5$ km, $\Delta L = -196.6$ km, $\Delta\bar{v} = -3.79$ m/s; 800 km: $\Delta d = +84.2$ km, $\Delta L = -233.3$ km, $\Delta\bar{v} = -4.18$ m/s. Here, positive distance and negative path-length differences mean that the 12 m/s policy is farther from the center and travels less; negative mean-speed differences reflect its lower propulsion use.

The relative time-averaged distance penalty of the 12 m/s policy is 100 km: +6.3%; 200 km: +10.6%; 400 km: +18.2%; 600 km: +17.7%; 800 km: +14.0%. The absolute difference therefore increases with deployment radius, although not monotonically in relative terms. This pattern is consistent with the finite time available to remove forecast-induced displacement. The shorter 12 m/s paths reduce propulsion demand, but the reduced speed is associated with weaker TC-center tracking in the tested episodes.

To provide an internally consistent visual comparison, Figs. 3–5 use the same held-out CHAMPI (2015) segment. The surface figures show motion from the segment start to its endpoint, while the subsurface rollout starts from the segment-end anchor, representing the sequential transition from surface delivery to underwater observation.

Figure 3 resolves the temporal evolution for the four-SUV, 400 km case. At deployment, the four vehicles are dispersed around the formation center far from the TC. During the first several hours, the vehicles turn toward the strengthening wind region and begin to reduce the initial displacement. By 6.5 h, all four vehicles have initiated the inward turn; by 13.0 h they have entered the outer high-wind region; and by 19.5 h all four trajectories have turned into the high-wind branch and substantially reduced their initial displacement. The individual paths differ because each vehicle has its own state and position within the shared fleet-level wind observation; nevertheless, the fleet-level motion follows the wind pattern toward the TC core and illustrates long-range error correction in this representative episode.

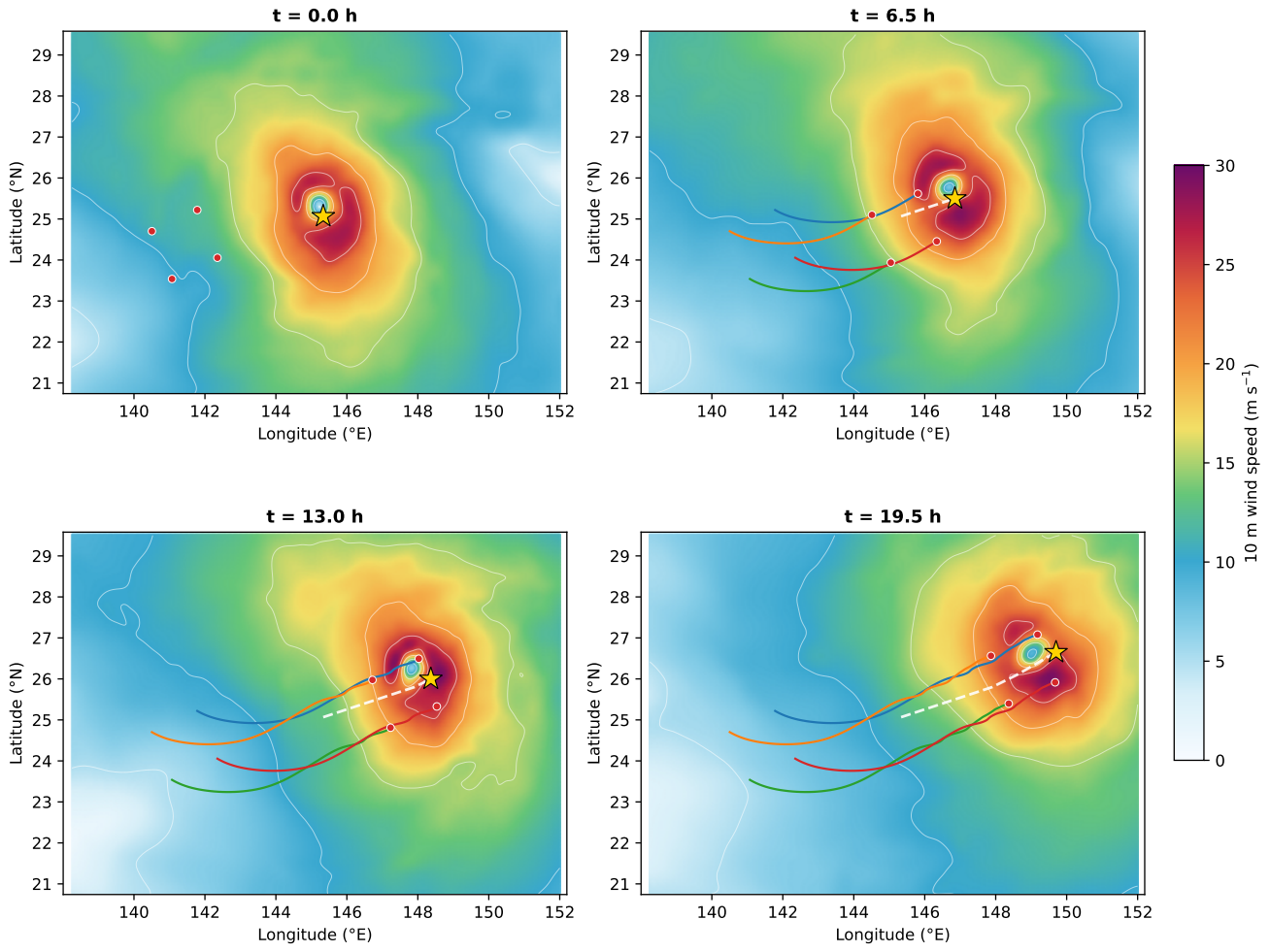


Figure 3: Four-SUV surface-tracking process for CHAMPI (2015) at the 20 m/s speed limit. The four panels show the initial deployment and subsequent states at approximately 6.5, 13.0, and 19.5 h. The color background is the spatially interpolated ERA5 10 m wind-speed field at each time. Small red dots mark vehicle positions, and the yellow star marks the TC center.

Figure 4 shows how initial forecast error changes the surface search process. At 100–200 km, the formations converge rapidly and remain concentrated around the TC core. As the deployment radius increases to 600–800 km, the trajectories lengthen and fan out over a wider region before turning toward the TC. Increasing the fleet from 4 to 16 SUVs produces a denser set of sampled branches and broader spatial coverage, but it does not remove the need for long-range transit. In the 12- and 16-SUV cases, many paths overlap near the core because the shared wind observations guide the vehicles toward the same high-wind region. In these representative cases, fleet size mainly broadens the sampled approach region, whereas deployment error primarily determines the required transit distance.

The 600 and 800 km rows also reveal two coupled failure mechanisms at large initial displacement. At 600 km, the representative fleet can still recover a usable inward direction from the observed wind-speed field and its spatial gradient, so the vehicles continue toward the TC despite the longer transit. At 800 km, the vehicles must first overcome a larger geometric separation with finite speed and a finite episode duration; in addition, the TC-induced wind speed and its local gradient become weaker and less distinctive at that range. The reduced signal quality makes the approximate bearing toward the TC more ambiguous, so the policy may approach more slowly, change direction repeatedly, or fail to recover the TC within the episode. These two effects are coupled: longer travel time increases the consequence of even a small directional error, while the weaker remote wind signature makes such an error more likely.

5.2. Execution-Layer Action Filtering

Table 3 reports the low-pass filtering ablation over the same trained actor, 768 held-out segments, and 10 min decision interval. Without filtering, the 4-, 8-, 12-, and 16-SUV configurations travel 1294.9–1331.5 km per vehicle, with mean speeds of 19.23–19.73 m/s and maximum speeds of

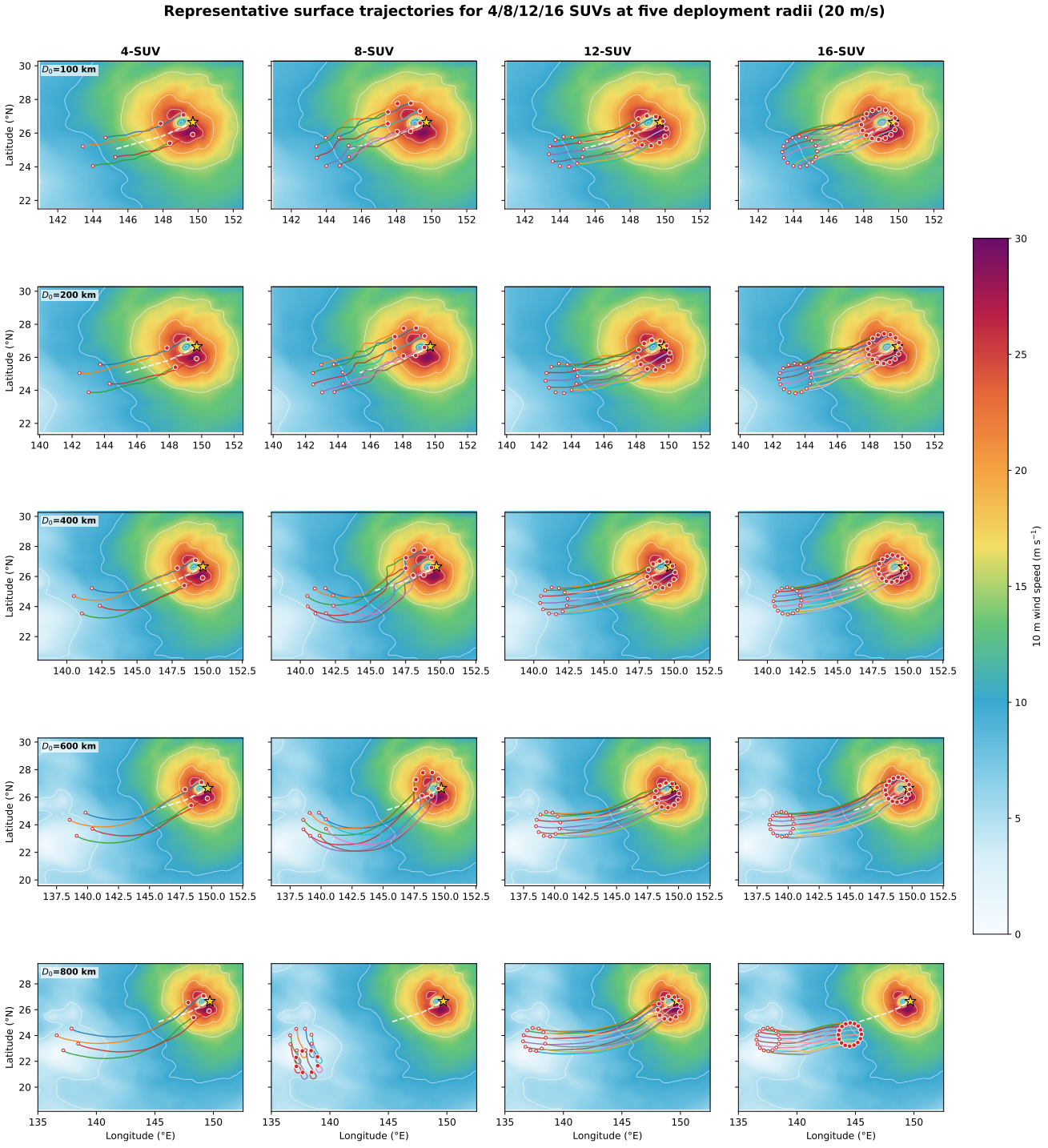


Figure 4: Representative CHAMPI (2015) trajectories for the final 20 m/s surface policy. Columns show 4-, 8-, 12-, and 16-SUV formations, while rows show nominal formation-center deployment radii of 100, 200, 400, 600, and 800 km. The color background is the spatially interpolated final-time ERA5 10 m wind-speed field. Thin colored lines are vehicle trajectories, open circles mark initial positions, small red dots mark final vehicle positions, and the yellow star marks the TC center at the end of the episode.

19.92–20.00 m/s. With the execution-layer filter ($\beta = 0.8$), path length falls to 690.8–769.8 km, mean speed to 11.11–12.72 m/s, and maximum speed to 16.13–18.20 m/s. The fleet-mean final SUV-to-TC-center distance changes from 288.2 to 258.2 km for four SUVs, from 305.2 to 308.6 km for eight SUVs, from 478.5 to 330.0 km for 12 SUVs, and from 313.6 to 273.4 km for 16 SUVs. Averaged over the four fleet sizes, the final distance decreases from 346.4 to 292.6 km, a reduction of 53.8 km (15.5%). Thus, filtering not only avoids degrading final tracking but improves it on average: three fleet sizes improve, whereas the eight-SUV case increases only slightly, by 3.4 km.

Table 3
Effect of execution-layer low-pass filtering for 4/8/12/16 SUVs on the 500-TC benchmark. The same trained policy is evaluated with the filter off and on ($\beta = 0.8$) over 768 held-out segments.

Fleet	Filter	Final mean distance (km)	Path per SUV (km)	Mean speed (m/s)	Maximum speed (m/s)
4	Off	288.2	1331.5	19.73	20.00
4	On	**258.2**	**768.1**	**12.72**	**18.20**
8	Off	**305.2**	1318.0	19.57	19.97
8	On	308.6	**770.9**	**12.58**	**17.68**
12	Off	478.5	1309.3	19.34	20.00
12	On	**330.0**	**690.8**	**11.11**	**16.13**
16	Off	313.6	1295.0	19.23	19.92
16	On	**273.4**	**721.4**	**11.94**	**16.98**

Final mean distance is the fleet average of the individual SUV-to-best-track-center distances at the final time step. Off/on changes only the execution-layer filter on the same trained actor.

The same comparison shows reductions for every fleet size: path length decreases by 563.4, 547.1, 618.5, and 573.6 km for 4, 8, 12, and 16 SUVs, respectively, while maximum speed decreases by 1.80, 2.29, 3.87, and 2.94 m/s. The 12-SUV configuration has the largest relative path-length reduction, although the effect is not monotonic in fleet size. These results show that the execution filter improves final tracking on average while substantially shortening trajectories and reducing mean and peak speed.

5.3. Speed-Limit and Deployment-Error Analysis

The 12 and 20 m/s cases answer different engineering questions. The 12 m/s value is close to the published surface-speed scale of the current Blue Whale, whereas 20 m/s represents a future high-speed scenario. A simple direct-pursuit estimate illustrates why deployment radius changes the effect of this choice. If the TC translates at speed U_c and the SUV moves toward it at V_{max}, the nominal closure rate is approximately

$$V_{close} = V_{max} - U_c. \tag{16}$$

For a deployment error D, the corresponding correction time is

$$t_{corr} \approx \frac{D}{V_{close}}. \tag{17}$$

This is only a kinematic lower bound because it neglects turning and imperfect center estimation, but it exposes the dominant scale. At 100–200 km, both speed limits provide enough closure to correct a moderate forecast displacement. At 600–800 km, a TC translation speed of 4–7 m/s removes a substantial fraction of the 12 m/s speed budget. The required correction time can then approach or exceed the 20 h mean segment duration. A 20 m/s vehicle retains a larger reserve for reducing initial center error, but its benefit is expected mainly at the larger radii; at short radii both cases already have sufficient speed.

The radius-dependent pattern reported in Section 5.1 is consistent with this closure-rate argument. At the 800 km setting, the mean time-averaged distances of the 20 m/s fleets are 585.6–617.0 km, compared with 658.2–701.0 km for the 12 m/s fleets. The 12 m/s policy instead follows a shorter path, saving 78.7–233.3 km per vehicle on average across the five radii and operating at a lower mean speed. This is a genuine efficiency–error trade-off: the 12 m/s scenario spends less propulsion effort but has less ability to recover a large forecast displacement error. The gap is relatively modest at 100–200 km, where both speed caps are adequate, and becomes operationally material at 400–800 km.

This mechanism determines how the deployment-error curves should be read. Differences between 12 and 20 m/s that grow with radius indicate insufficient relative speed for forecast-error correction, whereas a similar residual error at both speeds indicates that the limiting factor is observation information, formation geometry, or unresolved TC motion rather than the velocity cap. The 800 km protocol is therefore conservative for the 12 m/s platform: it includes errors that cannot always be fully removed within one segment. Operationally, that case argues for either a faster platform, a tighter track-forecast error envelope, or earlier deployment rather than relying on post-deployment speed alone.

5.4. Underwater Strategy Comparison

The final subsurface matrix uses four SUVs and 20 h episodes. It compares active IPPO, random control, and passive drift at ocean depths of 0, 10, 20, 30, 40, 50, and 60 m below the sea surface under maximum propulsion speeds of 1.5 and 2.0 m/s. Because the surface stage is responsible for reaching the TC core, the underwater stage assumes that this long-range correction has already been completed. Vehicle positions are initialized uniformly within 200 km of the TC center, and the 100/200/400/600/800 km test-radius protocol is not applied to the underwater policy. Passive drift exploits the TC-induced flow directly and expends no propulsion effort. Active IPPO and random control provide the two controlled alternatives against which the value of active propulsion is evaluated.

Unlike the surface stage, the underwater stage assumes no inter-vehicle communication. Each SUV conditions its action only on its own current speed, current components, five-frame current history, and propulsion velocity. It does not receive the observations, positions, or actions of the other

SUVs. The actor parameters are shared to ensure consistent learning across vehicles, but information is not exchanged during execution.

Under this communication-free and decentralized formulation, increasing the number of vehicles at a given depth would expand the aggregate sampling footprint but would not provide any individual policy with additional observations or decision information. The experiment was therefore designed to test the decision strategies rather than to estimate fleet scaling underwater. Fixing the fleet at four vehicles per depth also keeps the active-control, random, and passive-drift comparisons paired on the same episodes. In an actual 16-SUV deployment, the fleet can instead be divided into four groups of four, with each group assigned to a different ocean depth. Successive groups or deployments can then cover the seven depth layers without requiring more than four independently acting vehicles at any one layer.

The three strategies differ only in how the propulsion command is generated:

$$\begin{aligned} \mathbf{a}_i^{\text{active}} &= \text{clip}\left(\pi_\theta(\mathbf{o}_i), -1, 1\right), \\ \mathbf{a}_i^{\text{random}} &\sim \mathcal{U}\left([-1,1]^2\right), \\ \mathbf{a}_i^{\text{drift}} &= \mathbf{0}. \end{aligned} \tag{18}$$

During evaluation, the active IPPO policy uses the deterministic mean action, whereas random control samples a new independent two-dimensional action at every 10 min decision step. Active IPPO and random control use the same propulsion limit and equation of motion. The active policy conditions its action on the local current history, whereas random control ignores the observations. Passive drift applies zero propulsion and advances solely by ambient-current advection, so its trajectory also serves as the no-propulsion energy baseline.

For a vehicle i assigned to depth z, the ground path length after K decision steps is

$$L_i(z) = \sum_{k=0}^{K-1} \left\| \mathbf{x}_{i,k+1} - \mathbf{x}_{i,k} \right\|_2, \tag{19}$$

and the mean experienced current speed along that trajectory is

$$\bar{V}_i(z) = \frac{1}{K+1} \sum_{k=0}^{K} \left\| \mathbf{u}_{\text{ocean}}(\mathbf{x}_{i,k}, z, t_k) \right\|_2. \tag{20}$$

The figure reports the fleet-mean path length $N^{-1}\sum_{i=1}^{N} L_i(z)$ and the fleet-mean experienced current speed $N^{-1}\sum_{i=1}^{N} \bar{V}_i(z)$.

Figure 5 illustrates the qualitative trajectory construction for the three strategies. It is retained as a qualitative example only; all comparative inference is deferred to the final ten-seed, seven-depth evaluation.

Figure 6 defines the layout of the final seven-depth comparison and will be populated only after all ten-seed active policies have been retrained and evaluated under the corrected implementation. No quantitative underwater ranking is inferred from the present placeholder.

This behavior is consistent with the speed scales in the near-core region. For the paired 20 h episodes initialized within 200 km of the TC center, the area-averaged current speed at the evaluated 0, 30, and 50 m depths is 0.34–0.57 m/s, while the spatial 95th percentiles are 0.92–1.30 m/s and the 99th percentiles are 1.30–1.85 m/s. Strong local-current maxima reach 2.19–3.55 m/s across these depths, and the average local maximum within an instantaneous 200 km core area is 0.91–1.24 m/s. Thus, the high-current tail and local maxima are comparable to, or even exceed, the SUV propulsion limits of 1.5 and 2.0 m/s. This similarity in scales motivates the active-versus-drift comparison but does not by itself establish which strategy is preferable. Active propulsion can relocate a vehicle, whereas passive drift follows the ambient flow at zero propulsion effort; the final matrix will quantify the resulting differences in experienced current speed, coverage, ground path, and mean ground speed.

6. Conclusion

This paper presented a simulation-based multi-agent reinforcement learning (MARL) framework using IPPO for coordinated surface–subsurface observation of tropical cyclones with parameterized Blue Whale-class SUVs. Event-specific simulation environments were constructed from 500 historical TCs, with 450 TCs used for training and 50 for testing. ERA5 winds, HYCOM currents at seven depths from 0 to 60 m, and IBTrACS tracks were aligned into a common event-specific environment. Surface experiments covered 4-, 8-, 12-, and 16-SUV fleets, 12 and 20 m/s speed limits, ten seeds, and deployment errors from 100 to 800 km. Surface vehicles shared their wind observations across the fleet. After entering the 200 km near-core region, underwater vehicles used propulsion limits of 1.5 and 2.0 m/s without inter-vehicle communication.

The surface results support an asymmetric surface operating strategy: 20 m/s improved long-range error correction, whereas 12 m/s reduced path length and mean speed but left a larger residual tracking error. Low-pass filtering improved final tracking on average while further reducing path length and peak speed. These results also show the intended observation–planning coupling: surface measurements continuously alter the next motion command rather than only populating a post hoc map. The underwater evaluation is configured to compare active IPPO, random control, and passive drift across all seven depths under the corrected unit-disk action and causal five-frame observation implementation; final quantitative conclusions are deferred until this matrix is complete. The present evidence is simulation-based and uses a mission-level horizontal point-mass approximation; no field validation or operational forecast impact is claimed. Future work should couple the planner to calibrated six-degree-of-freedom vehicle dynamics and evaluate the resulting observations in air–sea coupled data assimilation.

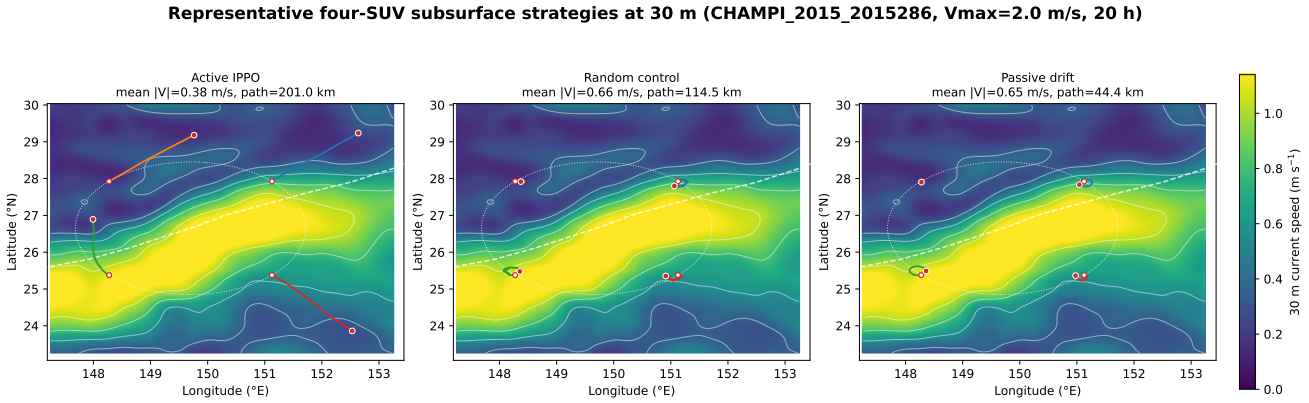


Figure 5: Representative four-SUV trajectories at 30 m depth for the same CHAMPI (2015) segment used in the surface trajectory figures. Rollouts start from the segment-end anchor, under Active IPPO, random control, and passive drift. The propulsion limit is 2.0 m/s and the episode duration is 20 h. Background color is the spatially interpolated episode-mean HYCOM current speed, dashed white curves show the TC best track, open circles mark the four vehicles deployed with 90° spacing on a 200 km initial ring, and small red dots mark final vehicle positions. This episode is illustrative and is not used for the final comparative statistics.

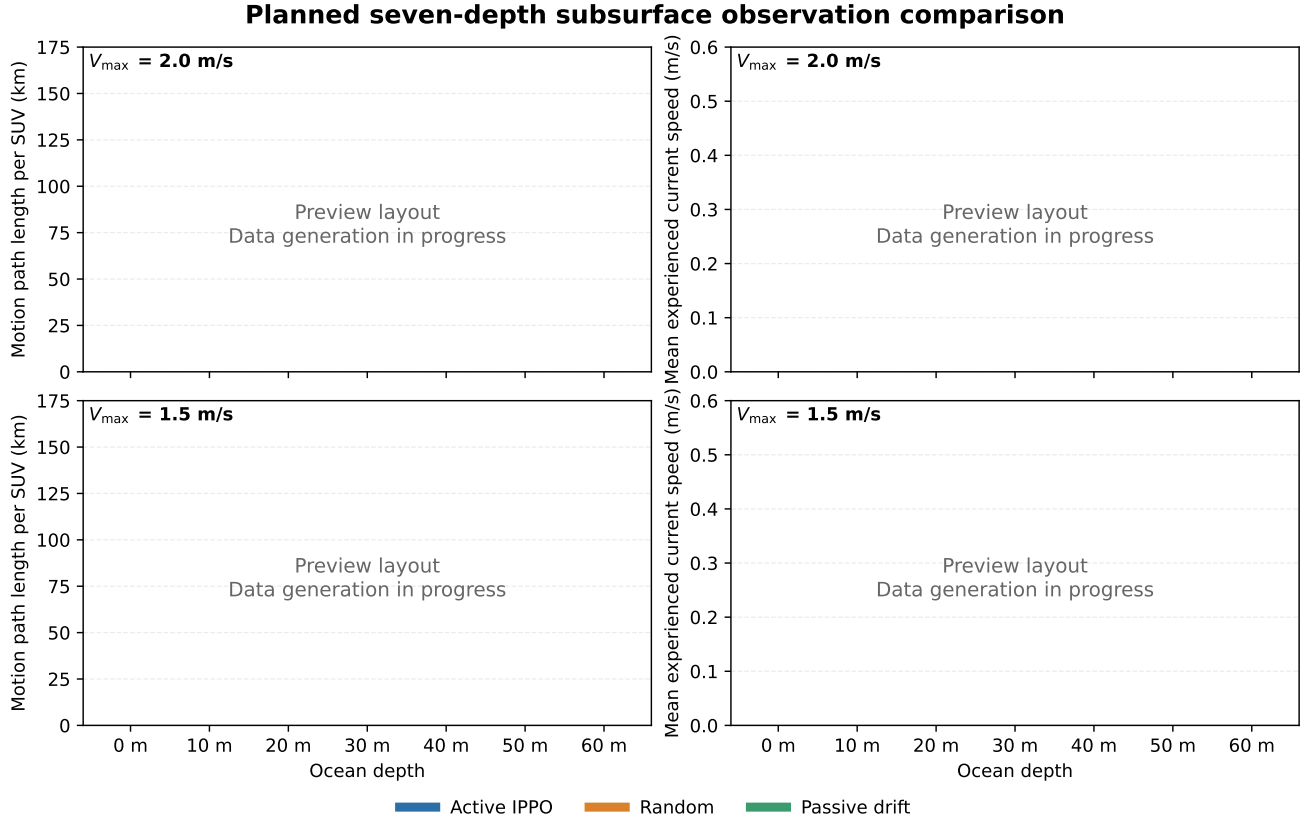


Figure 6: Preview of the planned seven-depth comparison at maximum propulsion speeds of 2.0 and 1.5 m/s. Columns show the fleet-mean motion path length and mean experienced current speed. Axes and legends define the final layout; no experimental values are plotted until all ten-seed seven-depth results are available.

Code and package availability

The final experiment code, simulation environments, evaluation scripts, and processed result tables are available at `https://github.com/DYHAI/SUV-MARL-TC`. The experiments used Python 3.11, Stable-Baselines3 2.9.0, Gymnasium 1.2.2, PyTorch 2.11.0+cu128, NumPy 2.4.6, pandas 2.3.3, xarray 2026.7.0, SciPy 1.17.1, and Matplotlib 3.11.1. Training and evaluation were performed on a workstation with a 24-core Intel Core Ultra 7 270K Plus CPU, 64 GB DDR5 memory, and an NVIDIA RTX 5060 Ti 16 GB GPU.

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A. Reward and IPPO Hyperparameters

Table A.1 summarizes the reward, action-filter, and safety parameters used in the surface and underwater experiments. Table A.2 lists the shared IPPO network and training hyperparameters.

Table A.1: Reward, action-filter, and safety parameters used in the final surface and underwater experiments.

Parameter	Symbol	Value	Description and applicable task
Panel A: surface reward and controller parameters			
Proximity-cost distance cap	$D_{\max}$	300 km	Distance cost is capped at 300 km to prevent excessively large gradients from distant initial deployments.
Near-center threshold	d_{near}	100 km	Threshold defining the discrete near-center bonus.
Near-center bonus	b_{near}	1.0	Bonus assigned when an SUV is within 100 km of the TC center.
Radius-excess threshold	R_{form}	100 km	Reference radius used by the additional distance penalty beyond 100 km.
Radius-excess weight	W_{form}	1.0	Weight of the penalty applied when an individual SUV is farther than 100 km from the TC center.
Action-change weight	λ_u	0.05	Weight of the squared action change $\|\mathbf{a}_{i,t} - \mathbf{a}_{i,t-1}\|_2^2$ at the 10 min interval.
Action-filter coefficient	β_{filt}	0.8	Coefficient of $\mathbf{a}_t^{\text{filt}} = (1-\beta)\mathbf{a}_t^{\text{raw}} + \beta\mathbf{a}_{t-1}^{\text{filt}}$. This is a controller setting rather than a reward term.
Surface reward clipping range	$[r_{\min}, r_{\max}]$	$[-5, 2]$	Clipping range applied to the team-averaged surface reward.
Panel B: underwater active-control comparison parameters			
Current-speed normalization	V_{norm}	1.0 m/s	Current-intensity reward is implemented as $\min(\|\mathbf{u}\|/V_{\text{norm}}, R_{\max})$.
Current-reward cap	$R_{\max}$	2.0	Upper bound of the normalized current-speed reward.
Propulsion-energy weight	λ_E	0.05	Weight of the propulsion penalty $\lambda_E\|\tilde{\mathbf{a}}_i\|_2^2$ in the current-seeking reward.
Downstream-alignment weight	$\beta_{\parallel}$	0.5	Weight of the positive propulsion–current alignment term.
Underwater reward clipping range	$[r_{\min}, r_{\max}]$	$[-5, 5]$	Clipping range applied to the team-averaged underwater reward during active control training.
Panel C: shared coastal-safety parameters			
Land penalty	P_{land}	20	Penalty applied when an SUV enters a land cell. Shared by the surface and underwater environments.
Nearshore buffer distance	D_{buf}	100 km	Width of the soft nearshore safety buffer.
Nearshore penalty weight	W_b	8.0	Maximum soft-penalty weight within the 100 km nearshore buffer.
Minimum deployment offshore distance	D_{deploy}	100 km	Minimum coastline distance required by offshore-only rejection sampling during deployment.
Action-filter safety margin	D_{filter}	100 km	Distance below which the onshore component of the commanded action is removed.

Note: the final subsurface comparison consists of active IPPO, random control, and passive drift. Passive drift does not optimize a learned propulsion policy or reward.

Table A.2: Parameter-sharing IPPO network and training hyperparameters used in the final surface and subsurface comparisons.

Parameter	Symbol	Value	Description
Algorithm	–	PPO	Stable-Baselines3 PPO with parameter sharing and independent per-agent action sampling.
Return aggregation	–	Team average	The environment reward is averaged across the N SUVs before computing the shared PPO return.
Optimizer	–	Adam	Adam optimizer used for the actor and critic parameters.
Policy network	–	2×256 MLP	Actor: Linear(d_{obs}, 256)–Tanh– Linear(256, 256)–Tanh– Linear(256, d_{act}).
Value network	–	2×256 MLP	Critic: Linear(d_{obs}, 256)–Tanh– Linear(256, 256)–Tanh– Linear(256, 1). Per-agent values are summed during training.
Per-agent observation dimension	d_{obs}	14/26/38/50	Surface wind-only observations for 4/8/12/16 SUVs, respectively. The four-SUV underwater active-control comparison uses $d_{\text{obs}} = 17$.
Surface observation	–	Wind at 10 m above sea level	Using ideal surface communication, each SUV receives the current wind speed and components at 10 m above the sea surface observed by every SUV in the fleet.
Underwater observation	–	Current only	With no inter-vehicle communication, the active IPPO controller observes only its own local current speed and components at the assigned ocean depth, together with a five-frame history and its own normalized propulsion velocity.
Action dimension	d_{act}	2 per SUV	Continuous east–north velocity command in $[-1, 1]^2$, followed by circular projection to enforce the maximum-speed constraint.
Parallel environments	N_{env}	4	Number of vectorized training environments.
Rollout length	n_{steps}	2048	Number of steps collected per environment before each PPO update.
Mini-batch size	B	256	Mini-batch size used during each gradient update.
Optimization epochs	N_{epochs}	10	Number of passes over each rollout.
Learning rate	α	3×10^{-4}	Adam learning rate used for the actor and critic.
Discount factor	γ	0.99	Reward discount factor.
GAE parameter	λ_{GAE}	0.95	Generalized advantage estimation applied to the team-average reward.
Policy clipping range	ϵ	0.2	Clipping range of the PPO probability ratio.
Value clipping range	ϵ_v	0.2	Clipping range of the value-function update.
Entropy coefficient	c_{ent}	0.01	Entropy regularization coefficient.
Value-loss coefficient	c_v	0.5	Weight of the value-function loss.
Maximum gradient norm	$g_{\max}$	0.5	Global gradient clipping threshold.
Training steps	T	2×10^6	Nominal environment steps per training run.
Independent seeds	–	0–9	Ten random seeds for each selected surface or active-control configuration.
Surface training forecast-error radius	R_{train}	0–800 km	The surface formation-center offset is sampled uniformly inside the 800 km disk around the initial TC center; both bearing and radial distance are randomized. This models the assumed upper bound of TC track-forecast error.
Surface validation center offsets	R_{test}	100/200/400/600/800 km	Fixed surface formation-center offsets with random bearings, evaluated without retraining. These values represent increasing track-forecast displacement.
Underwater initialization radius	$R_{0,\text{u}}$	$\leq$200 km	After surface convergence, subsurface vehicles are initialized uniformly within 200 km of the TC center. The 0–800 km forecast-error protocol is surface-only.
Decision interval	Δt	10 min	Environmental observation and action interval.
Surface formation radius	R_{ring}	100 km	Surface SUVs use equal angular spacing of $360°/N$ on a 100 km ring.
Surface fleet sizes	N	4/8/12/16	Number of SUVs in the final surface configurations.
Underwater fleet size	N	4	Number of SUVs in each active-IPPO, random-control, and passive-drift case.
Underwater depths below the sea surface	z	0/10/20/30/40/50/60 m	Seven ocean-current observation depths used in the final subsurface matrix.
Underwater methods	–	Active IPPO; random; drift	Three paired methods evaluated on the same episodes and deployments.
Underwater duration	H	20 h	Fixed mission window for the final subsurface comparison.
Surface speed limits	$V_{\max}^{\text{s}}$	12 or 20 m/s	Two surface-speed scenarios.
Submerged speed limits	$V_{\max}^{\text{u}}$	1.5 or 2.0 m/s	Two resultant propulsion-speed limits after unit-disk action projection, used for the active-control comparison.
Compute device	–	CPU	All reported PPO/IPPO experiments were trained on CPU.

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